Stall in Steady RANS Is a Branch-Selection Problem, Not a Convergence Problem

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Abstract

Two standard checks certify a steady RANS solution: grid convergence and iterative convergence. Using a 125-case campaign on the NACA 0012 (structured polar O-grids, $y^+ \approx 0.4$, k–ω SST with a γ–$Re_{\theta t}$ transition block), we show that at trailing-edge stall both checks pass while the answer is still wrong — because the discrete system possesses *multiple steady branches*, and neither refinement nor iteration selects among them. Under grid refinement at $\alpha = 15^\circ$ the upper-surface separation point jumps between three branches ($x/c = 0.53 \to 0.20 \to$ attached), while the same sequence is cleanly convergent at $\alpha = 5^\circ$ and single-branched at $\alpha = 18^\circ$. A doubled-horizon URANS verification ($t^* = 0.81$–2.4, 61 samples) then shows the converged steady state is not an attractor: windowed means drift monotonically away from it, and at $\alpha = 15^\circ$ the steady and time-averaged drags bracket the wind-tunnel measurement. The practical consequence is that at stall a single "converged" number is not a well-defined object; branch-resolved envelopes and horizon-dependence must be reported instead.

1 Introduction

For attached flows, the certification protocol of computational aerodynamics is settled: refine the grid until a functional changes by a tolerated fraction, iterate until the residuals stall, and report the result with a discretisation-error estimate. This letter asks what the same protocol certifies at *trailing-edge stall onset*. The evidence set is unusually dense: 125 scripted OpenFOAM runs, three-grid families at three incidences spanning the stall regime, and two time-accurate repeats with a doubled statistical horizon. The findings reduce to two statements: the steady solution set is not single-valued at stall onset (Section 3), and the selected branch is not an attractor of the unsteady system (Section 4).

2 Numerical experiment

Sections are meshed with Gmsh into structured polar O-grids (coarse 16,100, medium 33,740, fine 51,360 cells; $y^+ \approx 0.4$; farfield 20 chords, 50 for verification), closed with k–ω SST and, in a transition block, γ–$Re_{\theta t}$ [1]. Steady cases use `simpleFoam` (4,000 iterations; separation from skin-friction reversal). The URANS subset uses `pimpleFoam` at $Re = 3.6 \times 10^5$ [2] at $\alpha = 12^\circ$ and 15° on the 50-chord medium grid, sampled every 0.005 s. All 125 cases, meshes, and scripts are released for end-to-end reproduction.

3 Result 1: refinement does not select a branch at stall onset

Figure 1 tracks the upper-surface separation point through the refinement sequence (16,100 → 33,740 → 51,360 cells) at three incidences. At $\alpha = 5^\circ$ the flow is attached on every grid — the refinement ladder is boring, as it should be: lift changes by 0.58% and drag by 0.05% from medium to fine. At the post-stall angle $\alpha = 18^\circ$ both grids select the *same* deeply separated branch: separation $x/c = 0.17$ vs 0.15, drag within 3%. But at the stall-onset angle $\alpha = 15^\circ$ the three grids land on *three different branches*: $x/c = 0.53$ (coarse, $C_L = 1.12$), 0.20 (medium, $C_L = 0.99$), and fully attached (fine, $C_L = 1.24$) — a 26% lift spread across a monotone refinement sequence. No grid-convergence index can be formed from this family, and none should be: the family is not approaching a single limit, it is hopping between basins. The medium-grid production values reported across the campaign are therefore a *branch sample*, not a converged estimate — a distinction that only became visible because the family was run at the incidence where the physics switches.

4 Result 2: the selected branch is not an attractor

Suppose a branch is selected — is it at least the time-asymptotic state? The URANS repeats say no. Figure 2 shows sliding-window means (0.1 s window) of the force coefficients against convective time $t^* = tU_\infty/c$ over a doubled horizon ($t = 0.15$–0.45 s, $t^* = 0.81$–2.4, 61 samples; both runs required a time-step-halved restart after transition-correlation failures, and the histories are continuous through the restarts). Three facts stand out. First, the flow is violently unsteady:

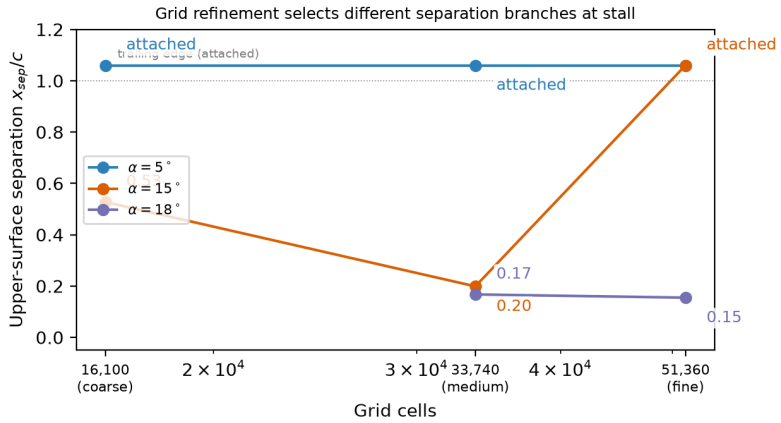


Figure 1: Upper-surface separation location through the refinement ladder. At $\alpha = 5^\circ$ (attached everywhere) and $\alpha = 18^\circ$ (one branch) refinement is well behaved; at stall onset ($\alpha = 15^\circ$) each grid selects a different separation branch.

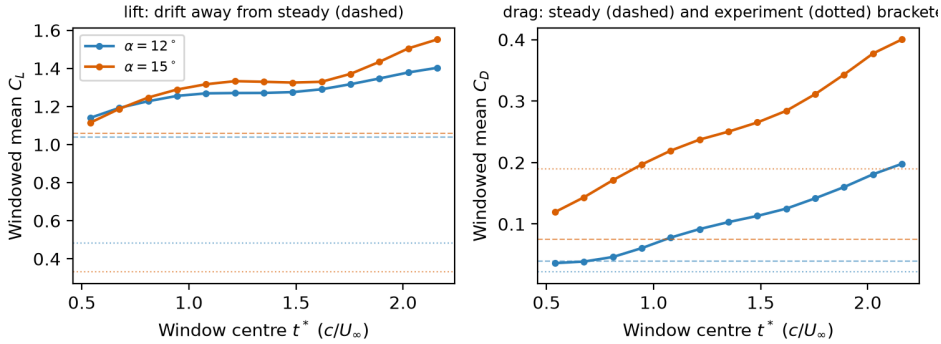


Figure 2: Sliding-window means over the doubled URANS horizon. Dashed: steady branch; dotted: Reynolds-matched experiment [2]. The means drift away from the steady values; at $\alpha = 15^\circ$ (drag, right) steady and URANS bracket the measurement.

the drag standard deviation is 24% ($\alpha = 15^\circ$) to 41% ($\alpha = 12^\circ$) of the mean. Second, the windowed means *drift* — at neither angle do they settle toward the steady branch; the mean lift excess over steady grows to 27% (1.322 ± 0.068 vs 1.039) and 34% (1.415 ± 0.108 vs 1.058) and is still growing at the end of the window. Third, at $\alpha = 15^\circ$ the steady drag (0.075) and the extended-window URANS mean (0.307) *bracket* the measured value (0.190): the earlier short-horizon mean (0.238 over $t^* = 0.81$–1.35) sat between the two. The horizon-averaged load of this non-stationary system is a function of the averaging window — so "the" URANS drag is as ill-defined as "the" steady drag.

5 What follows for practice

The failure is stall-specific, and that specificity is the useful part: attached-flow certification (the 0.58%/0.05% ladder) remains exactly as trustworthy as the protocol claims. At stall onset we suggest three reporting rules: (i) run the refinement family *at the stall angle* and report the branch map — a single-grid stall result is a draw from a discrete set, not an estimate; (ii) pair the steady solution with a time-accurate repeat, since the two can bracket the experiment from opposite sides; (iii) state the averaging horizon in convective units. None of this costs more than the standard protocol — it only places the family and the repeat where the physics dictates. The deeper limitation remains three-dimensional [3, 4]: the branch structure documented here is a property of the model, not of the airfoil.

6 Conclusion

At trailing-edge stall onset, steady 2D RANS fails in a specific, diagnosable way: refinement hops between separation branches, iteration cannot choose among them, and the chosen branch repels the time-accurate dynamics. "Converged" at stall is a statement about a branch and a window, not about a number. The 125-case evidence set, scripts, and diagnostics are openly released.

References

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